TWO GENUS-TWO SHIMURA CURVE QUOTIENTS, FROM THE PERIODS OF THEIR JACOBIANS

1. The curves

Let $D \in \{2002, 2030\}$ and let $X = X_0^D(1)^*$ be the quotient of the Shimura curve of discriminant D and level one by its full group of Atkin–Lehner involutions. Both have genus two. We exhibit models

$$X_0^{2002}(1)^* : \quad y^2 + (x^2 + x)\, y = 16x^6 + 4x^5 + 180x^4 - 456x^3 + 196x^2 + 64x + 4,$$
$$X_0^{2030}(1)^* : \quad y^2 + (x^2 + x)\, y = x^6 + 29x^5 + 13x^4 + 9x^3 + 159x^2 + 149x + 39,$$

and prove the arithmetic statements of Theorem 3 about them.

2. What is exact

Lemma 1. $\mathrm{Jac}(X)$ *is isogenous over* $\mathbb{Q}$ *to the abelian surface* A_f *attached to the newform orbit* f *of level* D*, weight two and trivial character whose Atkin–Lehner eigenvalue is* -1 *at every prime dividing* D*; that orbit is unique and of dimension two. For* $D = 2002$ *it is* `2002.2.a.f`*, with real multiplication by* $\mathbb{Q}(\sqrt{5})$*, and for* $D = 2030$ *it is* `2030.2.a.g`*, likewise with* $\mathbb{Q}(\sqrt{5})$*.*

Proof. By Jacquet–Langlands the space of weight-two forms on $X_0^D(1)$ corresponds to the subspace of $S_2(D)$ new at every prime dividing D, which at level one is all of $S_2(D)^{\mathrm{new}}$, and the correspondence reverses the Atkin–Lehner sign at each prime dividing the discriminant. The differentials on the quotient X are those fixed by every Atkin–Lehner involution, so they correspond to the newforms whose classical sign is -1 at each $p \mid D$. A search over the newforms of level D shows this condition picks out a single Galois orbit, of dimension two. The claim then follows from Eichler–Shimura. The orbit's identity and its coefficient field are read off the tables [2]; independently, the real multiplication field is the quadratic resolvent of the characteristic polynomial of Frobenius computed from the Eichler–Selberg trace formula, and agrees. □

Lemma 2. $\mathrm{Jac}(X)(\mathbb{Q})$ *is torsion free, for both* D*.*

Proof. For an abelian variety over $\mathbb{Q}$ with good reduction at an odd prime p, reduction is injective on rational torsion [1, Appendix], so $\#\,\mathrm{Jac}(X)(\mathbb{Q})_{\mathrm{tors}}$ divides $P_p(1)$ for the numerator P_p of the zeta function of X over $\mathbb{F}_p$. The point counts come from the trace formula, so P_p needs no model of X; over ten primes $p \le 43$ of good reduction the greatest common divisor of the $P_p(1)$ is 1. □

3. The construction

This section is numerical and is recorded for reproducibility, not as proof. Let Π be the period matrix of A_f, computed from the q-expansion of f. An integral alternating form Q on the period lattice satisfies Riemann's first relation $\Pi Q \Pi^t = 0$ exactly when it is a Néron–Severi class; for a surface with real multiplication those form a group of rank two, found by lattice reduction, and the principal polarizations among them are the classes of Pfaffian ± 1

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satisfying Riemann's positivity condition. Each such class gives a Riemann matrix τ, hence Rosenhain and then absolute invariants, which are recognised as rationals and fed to Mestre's construction. Several principal polarizations occur and they give different curves; the selection among them is made in Section 4 and not here.

4. What the point counts prove

Theorem 3. *Let C be either curve displayed in Section 1, and let f be the corresponding newform orbit of Lemma 1. Then C is a smooth projective genus-two curve over $\mathbb{Q}$; its discriminant is supported exactly on the primes dividing D; and for every prime $p < 60$ not dividing D,*

$$\#C(\mathbb{F}_p) \;=\; p + 1 - \operatorname{Tr}_{K/\mathbb{Q}} a_p(f),$$

thirteen primes in each case. Consequently $\operatorname{Jac}(C)$ *and A_f have the same characteristic polynomial of Frobenius at each of those primes.*

Proof. Each assertion is a finite exact computation with the displayed integral model: the genus and the discriminant from the model itself, the point counts by enumeration over $\mathbb{F}_p$, and $\operatorname{Tr} a_p(f)$ from the Hecke action on the modular symbols attached to f. The last sentence is the Eichler–Shimura relation, both sides being $p+1$ minus the trace of Frobenius. □

Remark 4. Theorem 3 does *not* prove $C \cong X$. What it gives is that $\operatorname{Jac}(C)$ and $\operatorname{Jac}(X)$ have the same Frobenius characteristic polynomial at thirteen primes. To conclude an isomorphism one needs, first, that this forces $\operatorname{Jac}(C) \sim A_f$ — an isogeny statement that would follow from the Faltings–Serre method with an explicit prime bound, which we have not carried out — and then that the isogeny can be upgraded to an isomorphism of principally polarized varieties, after which Torelli gives $C \cong X$ over $\overline{\mathbb{Q}}$ and the counts themselves fix the quadratic twist. Worse, the counts cannot even distinguish C from the other curves whose Jacobians are isomorphic to A_f as unpolarized varieties: those share an L-function, so they agree in $\#C(\mathbb{F}_p)$ at every prime. This is not hypothetical. At level 35, where an independent model of $X_0^{35}(1)/w_5$ is available from the method of Guo and Yang, the construction of Section 3 returns a curve with different absolute invariants — hence not isomorphic even over $\overline{\mathbb{Q}}$ — whose point counts nevertheless agree with it at all eight primes tested. So the agreement recorded in Theorem 3 is evidence for the isogeny class of $\operatorname{Jac}(C)$ and for the quadratic twist, and for nothing more; the curves displayed in Section 1 are candidates for the stars, not identifications of them.

Remark 5. The two facts in Section 2 are worth separating from the rest because they are unconditional and were obtained with no model of X in hand: Lemma 1 identifies the Jacobian up to isogeny, and Lemma 2 shows there is no rational torsion. The second is what makes a Borcherds divisor class on X vanish rather than merely be torsion, and so is used elsewhere independently of any model.

References

[1] N. M. Katz, *Galois properties of torsion points on abelian varieties*, Invent. Math. **62** (1981), 481–502.
[2] The LMFDB Collaboration, *The L-functions and modular forms database*, https://www.lmfdb.org.
[3] J. González, J. Guàrdia and V. Rotger, *Abelian surfaces of* GL_2*-type as Jacobians of curves*, Acta Arith. **116** (2005), 263–287.
[4] J.-F. Mestre, *Construction de courbes de genre* 2 *à partir de leurs modules*, in: Effective methods in algebraic geometry, Progr. Math. **94**, Birkhäuser, 1991, 313–334.